CPH 200C PS3 · REGIONAL AI · 2026

Three Governing Systems for AI

How the US, Europe, and Asia deploy AI: policy, funding, technical developments, and landscape by region

Curtis Chambers · Jinchi Wei

CPH 200C · Computational Precision Health

2026-06-04

FRAMING

Three governing systems for AI

Three regions, three governing systems for AI

UNITED STATES	EUROPE / EU	ASIA
Market-led	**Comprehensive**	**China-led**
Light federal + state patchwork; deregulate to 'win the race'. Closed frontier labs, deepest capital.	Horizontal risk-tiered AI Act (now softening). Sovereignty money, little frontier building.	State-driven core (China: open-weight, population scale). Korea/Japan/India build their own models. Asia owns the AI hardware: TSMC, HBM, Foxconn.

Same technology, three different games. The rest of the deck is this slide, scope by scope.

Regulate, build, deploy

	UNITED STATES	EUROPE / EU	ASIA
REGULATE	Light, late, contested	Heaviest, first-mover	State-led to light (varies)
BUILD	Frontier leader	ASML chokepoint only	Fast follower + owns hardware
DEPLOY	Enterprise SaaS	Compliance-gated, slow	Population scale

What leaders predicted, and what's different now

CHINA	UNITED STATES	EUROPE / EU
Kai-Fu Lee	**Eric Schmidt et al.**	**Mario Draghi**
AI Superpowers (2018)	The Age of AI (2021); NSCAI	EU Competitiveness Report (2024)
The age of implementation: execution and data, not breakthroughs, decide the race, favoring China.	AI is a national-security race the US must win; warned of China closing the gap fast.	Europe regulates but does not build; structurally behind on compute, capital, and scale.

WHAT'S DIFFERENT NOW (2025):

None predicted a Chinese OPEN model (DeepSeek) would reset the frontier while US labs re-closed. Lee's implementation thesis holds, but the ROI gap shows capability is not yet value.

POLICY & REGULATION

United States: executive whiplash plus a state patchwork

Executive whiplash	Trump rescinded Biden EO 14110 (Jan 2025); the July 2025 AI Action Plan deregulates to win the race; a Dec 2025 EO directs agencies to preempt state laws.
State patchwork	Colorado AI Act, Texas TRAIGA (Jan 2026), California SB 53, New York RAISE Act. A federal moratorium on state AI laws failed in the Senate 99 to 1.
Anti-woke procurement	EO 14319 bars federal purchase of LLMs that fail an ideological-neutrality test.
Frontier safety (Jun 2026)	Voluntary EO: labs grant up to 30-day pre-release access to NSA-designated frontier models + a Treasury cyber clearinghouse. No mandatory licensing, security-framed, not clinical.
Net effect	Light federal touch, a thickening state patchwork, and an open fight over who governs AI.

European Union: the AI Act, now softening

The AI Act	In force Aug 2024, phased: prohibited practices Feb 2025, foundation-model rules Aug 2025, high-risk 2026 to 2027. Penalties up to 35M euros or 7 percent of global turnover.
The softening	The Nov 2025 Digital Omnibus delays high-risk rules to 2027 and 2028 under competitiveness pressure; the separate AI Liability Directive was scrapped.
Regulates, does not build	Only 11 percent of EU firms use AI; 73 percent of foundation models are US-origin. Draghi: it is too late to build systemic challengers to US cloud.
Sovereignty money	InvestAI aims to mobilize 200B euros and build AI gigafactories, but ~90 percent of EU digital infrastructure is foreign-controlled.

China: vertical rules and alignment to state values

Vertical, not omnibus	No single AI law; iterative ministry rules. The 2023 Interim Measures govern public GenAI; 748 services were filed with regulators by Dec 2025.
Mandatory labeling	Since Sept 1 2025, all AI-generated content must carry explicit and hidden (metadata) labels.
Aligned to state values	Models must uphold core socialist values; China dropped a comprehensive AI law from its 2025 agenda in favor of pilots and standards.
Governance offensive	At the 2025 World AI Conference, Beijing proposed a World AI Cooperation Organization, courting the Global South against US go-it-alone policy.

Asia’s regulatory split

China	State-vertical	Iterative ministry rules, AI-content labeling, alignment to state values. No omnibus law.
South Korea	Comprehensive	AI Basic Act (effective Jan 2026), the 2nd after the EU, but far lighter (fines cap ~$21k).
Japan	Innovation-first	AI Promotion Act (2025): no penalties, guidance and name-and-shame only.
India	Light-touch	IndiaAI Governance Guidelines (Nov 2025); no standalone AI law, existing statutes adapted.
Singapore	Voluntary + tooling	Model AI Governance Framework + AI Verify, the first government AI-testing toolkit.

Regulatory philosophy, side by side

LIGHT-TOUCH	COMPREHENSIVE	STATE-VERTICAL
US, UK, Japan, India	**EU, South Korea**	**China**
Innovation first. Few or no penalties. Rules follow harm, if at all.	One horizontal law, risk-tiered, ex-ante. Heaviest compliance load.	Iterative sector rules + industrial policy + alignment to state values.

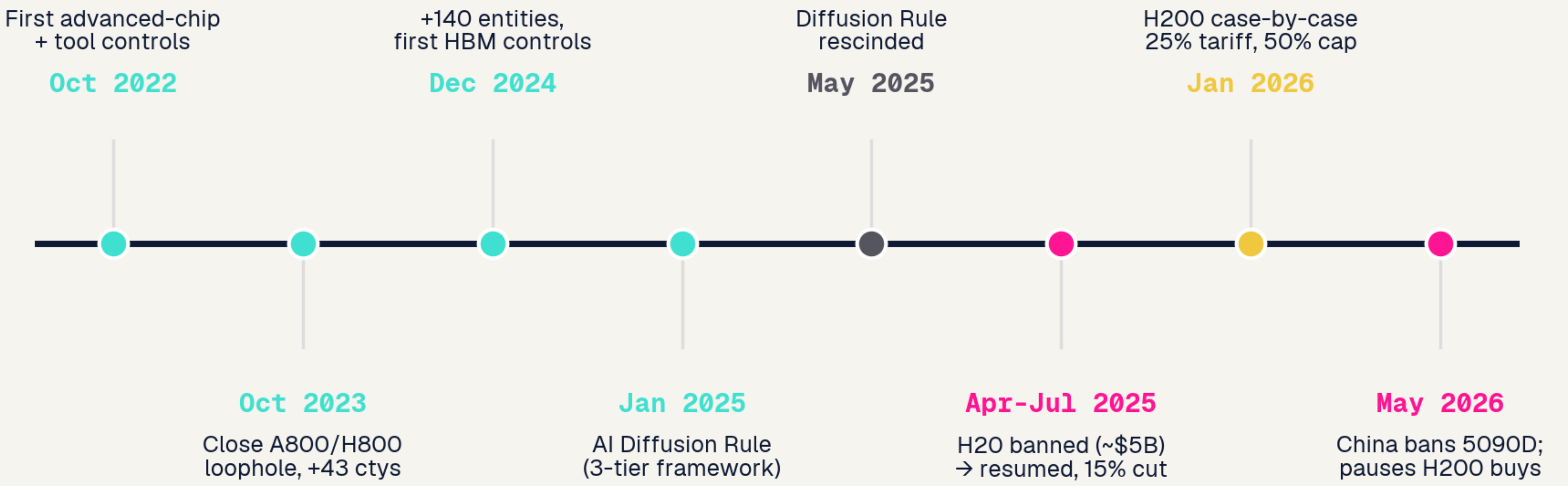
ASIA SPLITS

Japan: AI Promotion Act (2025), no penalties. South Korea: AI Basic Act (effective Jan 2026), EU-style comprehensive. India: light-touch + the IndiaAI Mission.

GEOPOLITICS & COMPUTE

The chip war, 2022 to 2026

The chip war, 2022-2026: escalation, reversal, and China's pivot to domestic silicon



Sources: BIS / Commerce rules, NVIDIA filings, Tom's Hardware. The US tightened for three years, abruptly reversed in 2025, and China decoupled regardless.

The NVIDIA saga and China's domestic stack

EXPORT CONTROLS + NVIDIA

- H20 banned Apr 2025 (~$5B NVIDIA charge), resumed Jul 2025 with a 15% cut of China revenue to the US government.
- H200 moved to case-by-case Jan 2026: 25% tariff, 50% volume cap, US testing.
- NVIDIA China share fell below 60%; the House AI Overwatch Act would bar Blackwell for two years.

CHINA'S DOMESTIC STACK

- Huawei Ascend 910C ~60% of an H100; CloudMatrix 384 links 384 chips to rival NVIDIA racks.
- Nine domestic chips put on the secure-and-reliable procurement list; state data centers must buy local.
- Chinese firms shipped 1.65M AI GPUs in 2025; China banned the 5090D and paused H200 buys.

Scale versus constraint: two paths to better models

UNITED STATES: SCALE

Answer the race with compute: Stargate ($500B), ~$700B hyperscaler capex in 2026, nuclear power deals. Frontier models stay closed. Abundant GPUs mean less pressure to economize.

Stargate; hyperscaler capex 2026

CHINA: CONSTRAINT

Capped on advanced GPUs by export controls, efficiency becomes the edge: mixture-of-experts, distillation, DeepSeek's ~$6M training run, cheap inference. Necessity drives method innovation.

DeepSeek-V3 technical report, 2025

EUROPE: the upstream chokepoint. ASML's EUV monopoly is the gate the controls run through.

The provocative read: the GPU ceiling may be pushing China toward genuine methodology innovation, while the US mostly scales hardware. May 2026: Huawei's chairman publicly thanked the US for export controls, saying they 'supercharged' China's chip industry (Tom's Hardware; SCMP).

Compute and energy: the binding constraints

UNITED STATES

16 GW

data-center pipeline, but only ~5 GW under construction. ~$700B hyperscaler capex in 2026. Power is the bottleneck.

CHINA

~400 GW

projected spare power by 2030 (added 430 GW wind+solar in 2025). 40 GW DC by end-2026. Energy is the advantage.

EUROPE / EU

EUR 200B

InvestAI + AI gigafactories. ASML holds the EUV monopoly. Compute and capital are the gap.

Asia owns the AI hardware

TSMC

Taiwan

Fabricates essentially all leading-edge AI chips. NVIDIA booked >60% of its CoWoS advanced-packaging capacity.

HBM

South Korea

SK Hynix holds ~62% of HBM memory and is NVIDIA's lead supplier; with Samsung, a global duopoly.

Foxconn

Taiwan

Builds >40% of the world's AI servers, the lead integrator for NVIDIA GB200/GB300 racks.

The hedge: sovereign AI means local-language models on foreign chips. Singapore runs Qwen; Malaysia runs DeepSeek on Huawei. Asia makes the hardware the world's AI runs on, but rents the models.

CORPORATE & MODELS

United States: the closed frontier and a capital arms race

$5.2T	$852B	~$965B	$700B
NVIDIA market cap (world #1)	OpenAI valuation	Anthropic (press-reported)	2026 hyperscaler capex

The frontier shifted back to CLOSED: after Llama 4 stumbled and DeepSeek leveraged its open weights, even Meta pivoted toward proprietary models. OpenAI, Anthropic, and Google keep their best models closed to recoup training cost and limit misuse.

Europe: one chokepoint, a thin application layer

ASML

The one irreplaceable asset. Monopoly on EUV lithography (~90% share, ~$250M/machine); EUR 32.7B 2025 sales. Every advanced NVIDIA or TSMC chip depends on it. Europe's leverage is hardware, not models.

Mistral (FR)

EUR 11.7B; ASML took ~11%

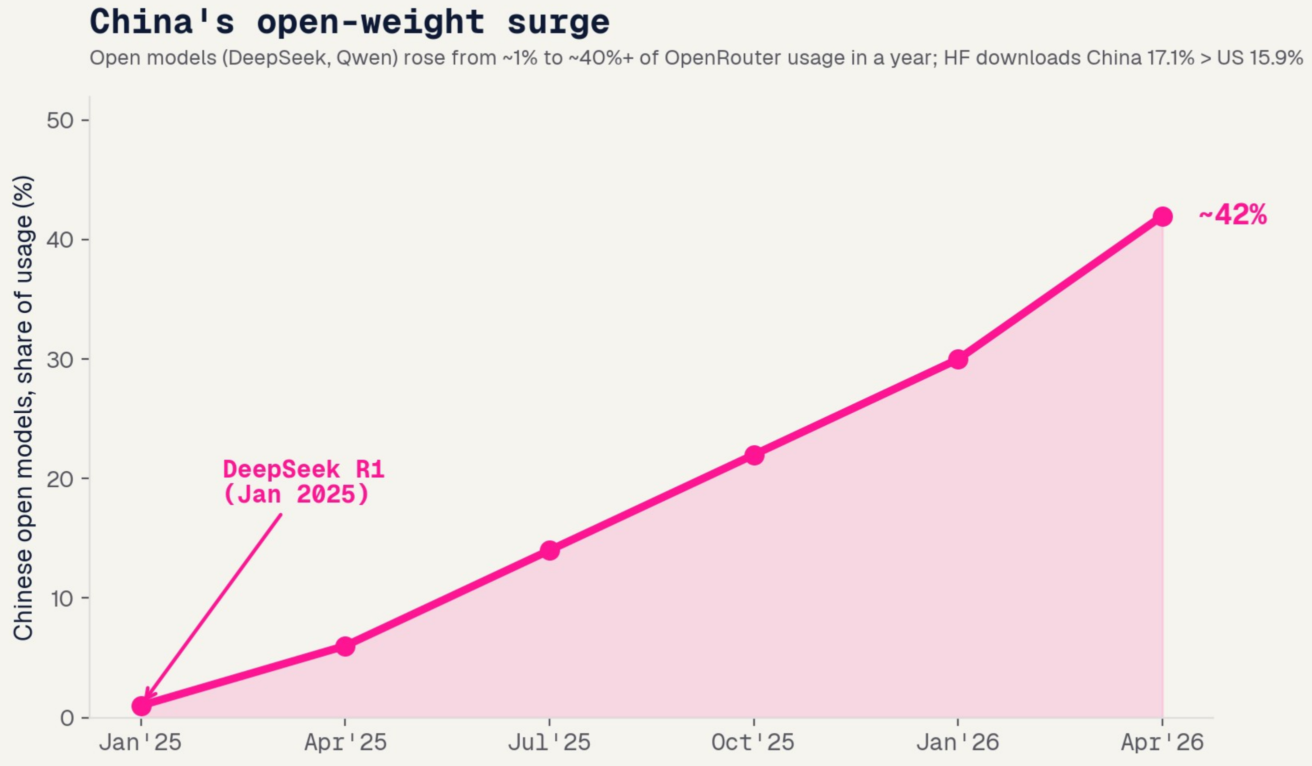
Helsing (DE)

defense AI, EUR 12B

ElevenLabs / Synthesia

$11B / ~$4B; application layer

China: the open-weight surge



The DeepSeek moment

R1 (Jan 2025), trained for a reported ~$5.6M, topped the US App Store and wiped ~$600B off NVIDIA in a day.

An open-weight ecosystem

Qwen, Ernie, Kimi, GLM, Doubao, Pangu. Price wars (Qwen cut up to 97%); Doubao serves >120T tokens/day.

Global South adoption

Singapore's government chose Qwen; Malaysia's stack runs on DeepSeek. Open weights are becoming an export.

Asia beyond China

INDIA

POPULATION-SCALE

Sarvam (govt-backed sovereign LLM) + Krutrim, under the national IndiaAI Mission. AI layered on UPI / Aadhaar at population scale; Qure.ai screens TB nationally.

IndiaAI Mission; PIB 2025

SOUTH KOREA

CLINICAL-AI EXPORTER

Naver HyperCLOVA X, LG EXAONE (open-weight), Samsung Gauss, backed by a national AI fund. Lunit + VUNO export FDA-cleared clinical AI worldwide.

MSIT; TechCrunch 2025

JAPAN

INNOVATION-FIRST

Sakana AI, the country's top unicorn; SoftBank owns ~11% of OpenAI. No-penalty AI law; AI-hospital pilots; Rapidus chasing a 2nm fab.

AI Promotion Act 2025

SOUTHEAST ASIA

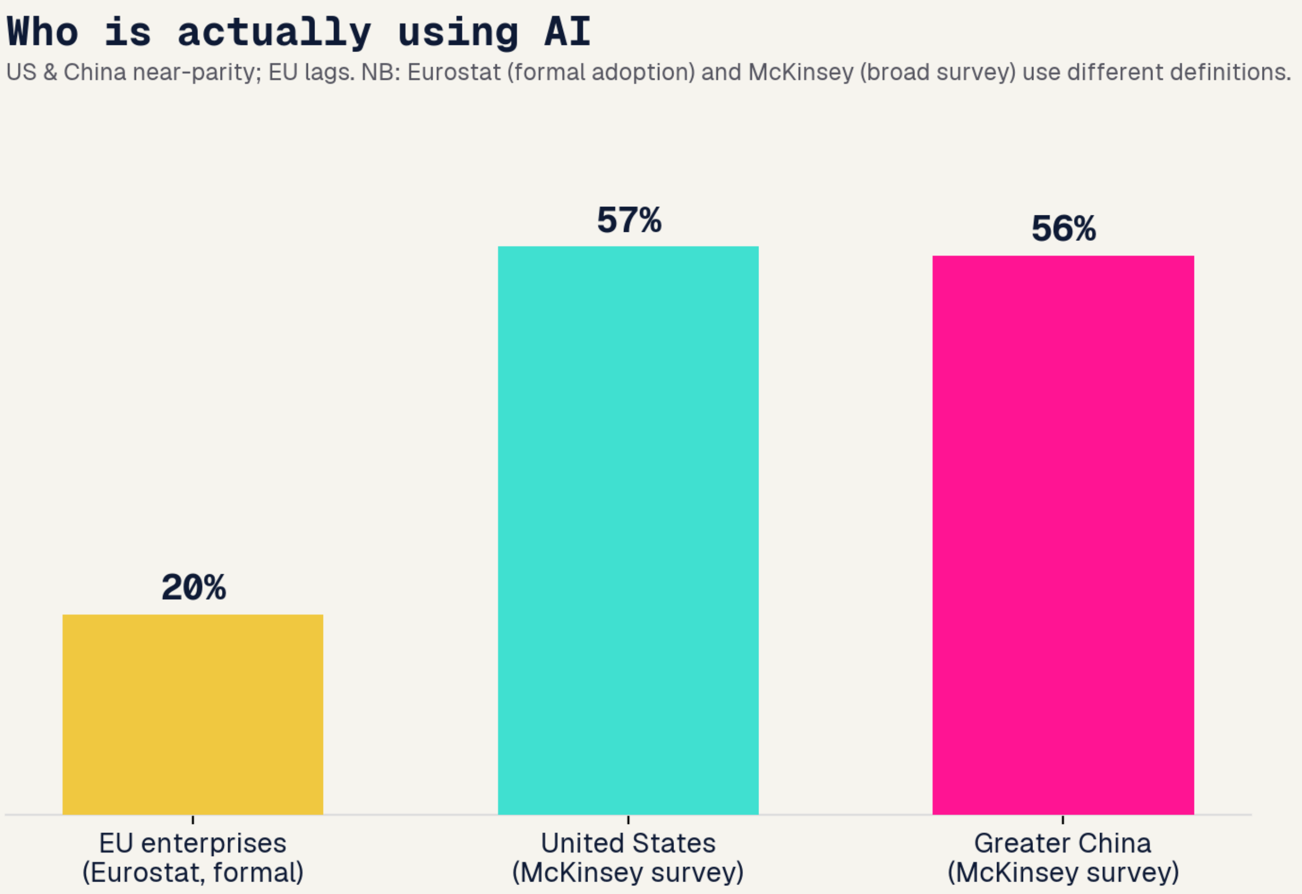
THE BATTLEGROUND

SEA-LION switched from Meta's Llama to Alibaba's Qwen; Sahabat-AI (Indonesia), FPT (Vietnam). US and Chinese models compete for the next billion users.

TechNode 2025

DEPLOYMENT & ADOPTION

Who actually uses AI



Eurostat measures formal adoption; McKinsey is a broad survey. Different definitions, same conclusion: the EU lags.

Near-parity on use

McKinsey: 88% of organizations use AI somewhere, but only ~1/3 have scaled it and 39% see real EBIT impact.

Consumer

ChatGPT ~900M weekly users, ~2.5x Gemini. China runs on its own: Doubao ~155M, DeepSeek ~82M. Western apps are absent there.

How AI gets deployed

CHINA

Super-app + state

AI agents inside WeChat; agentic commerce; state-coordinated, hardware-integrated. Deployment is centralized and consumer-facing.

UNITED STATES

Enterprise SaaS

AI flows through the vendor layer (Microsoft, Google, Salesforce). Specialized-vendor buys succeed ~67% vs ~1/3 for internal builds.

EUROPE / EU

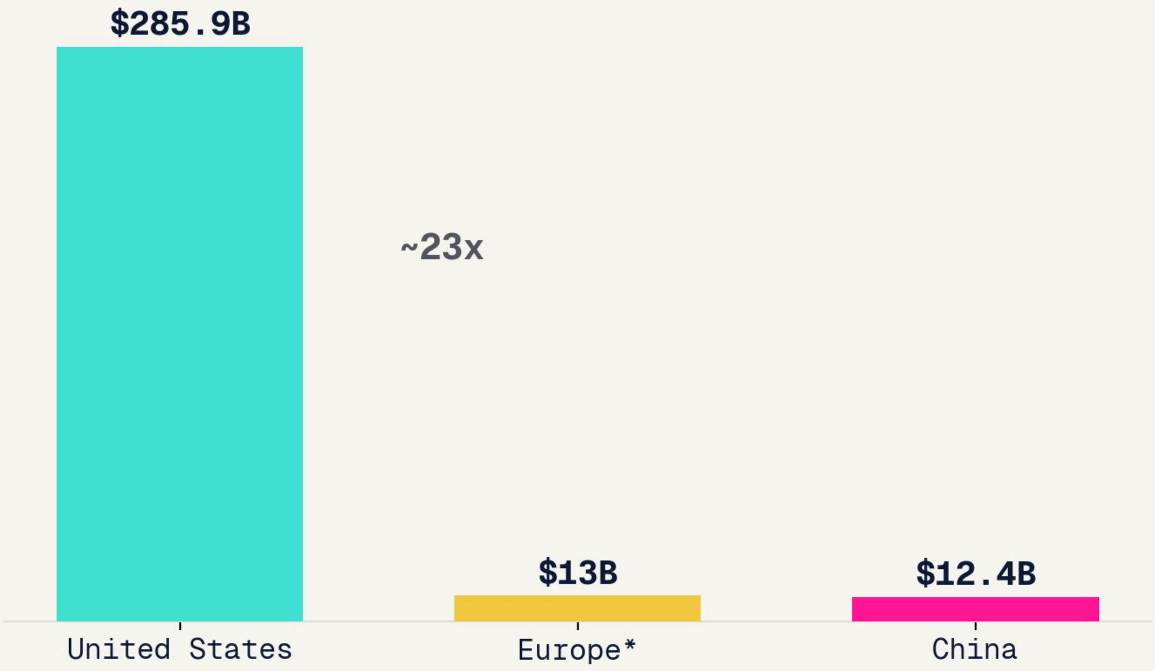
Compliance-gated

~60% of EU/UK developers report AI Act launch delays; deployment is slowed by the rulebook it wrote.

The capability scoreboard and the ROI gap

Private AI investment, 2025

Stanford AI Index (US, China). *Europe = est. EU+UK AI VC. The US captured ~83% of global private AI investment.



Models (2025)

US ~50 / China ~30 notable models

Frontier

Opus 4.8 / GPT-5.5 / Gemini 3.1 lead; DeepSeek V4 near-frontier at ~1/10 cost (US still on top)

Patents

China holds ~70% of granted AI patents

Investment

US ~83% of global private AI funding

The ROI gap: MIT found 95% of enterprise generative-AI pilots delivered no measurable P&L impact, despite $30-40B spent. Capability and capital are not the same as realized value.

STARTUPS & SUPPORT

Capital and leniency

$222B

US AI startup funding (65% of all US VC)

state

China private VC fell <25%; guidance funds filled in (140+ gov deals)

gap

Europe: structural late-stage capital gap

LENIENCY, most permissive to ship fast:

US	China	UK	EU

permissive ---→ constraining (EU sandboxes now delayed to Aug 2027)

Government support

UNITED STATES

$500B Stargate (privately financed) + CHIPS Act spillover. Light direct funding, heavy private capital.

EUROPE / EU

InvestAI EUR 200B; France alone pledged EUR 109B at the Paris summit, the largest single-event government commitment.

UNITED KINGDOM

AI Growth Zones, a Sovereign AI Unit up to GBP 500M, and a pledge to grow sovereign compute 20x by 2030.

CHINA

National VC Guidance Fund (~$143B over 20 yrs); the guidance-fund system runs ~2,178 funds worth ~$900B, plus procurement preference.

Talent, immigration, and founder-friendliness

UNITED STATES: drawbridge UP

A $100,000 per-petition H-1B fee took effect Sept 2025, a startup tax. ~60% of top US AI startups had an immigrant founder; ~70% arrived on student visas.

CHINA: drawbridge DOWN

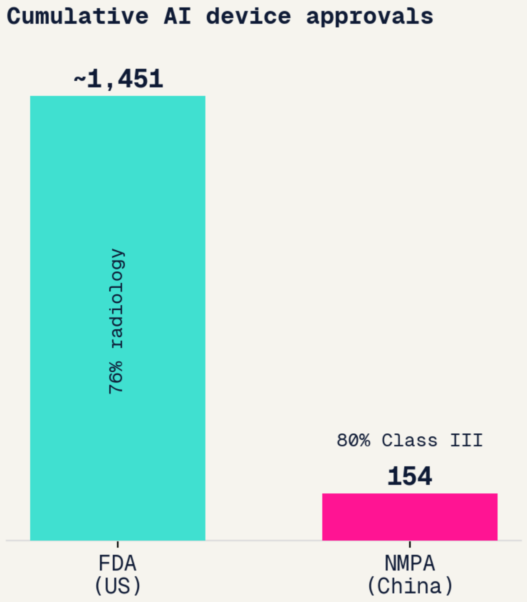
The new K-visa (Oct 2025) is uncapped, 5-year, no employer sponsor, aimed at STEM/AI. Returnee sea-turtles anchor frontier labs, timed to exploit US restrictions.

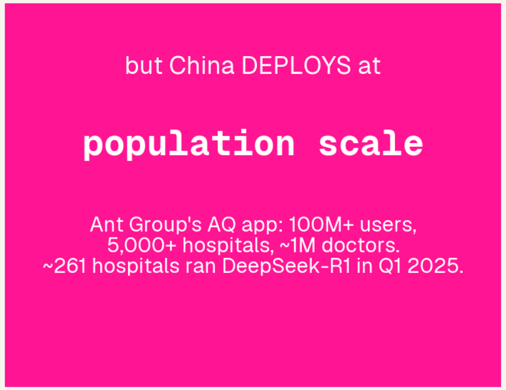
Founder-friendliness: US = capital + market, rising talent cost. China = state capital + deployment speed, but directed + walled. EU = subsidy + legal certainty, heaviest compliance. UK = most balanced.

THE HEALTH THREAD

Clinical AI by region

Clinical AI: the US approves the most discrete tools; China deploys at the most scale





United States

~1,451 AI devices cleared, 76% radiology, but reimbursement-starved and no pathway for generative/LLM clinical AI.

Deployed: Aidoc (1,600+ centers, 100M+ cases), Viz.ai stroke (1,800 hospitals).

Europe / EU

Medical AI is high-risk under the AI Act, stacked on MDR/IVDR, a double conformity burden; the health-data space is deferred to 2029.

Deployed: Oxipit ChestLink, world-first fully autonomous radiology AI (CE IIb).

China

Fewer approvals (154, 80% Class III) but deploys at population scale, state-coordinated.

Deployed: Infervision ~20k lung scans/day; Ant AQ 100M users; DeepSeek-R1 in 261 hospitals.

The regulate, build, deploy split, in medicine

UNITED STATES	EUROPE / EU	CHINA
APPROVES most	**REGULATES hardest**	**DEPLOYS most**
~1,451 discrete tools, device-by-device 510(k). But reimbursement is the bottleneck and LLM clinical AI has no pathway.	AI Act + MDR/IVDR double conformity, notified-body bottleneck. Germany's DiGA is the clearest working reimbursement route.	Population-scale platforms (AQ 100M users); 261 hospitals ran DeepSeek-R1. State coordination + volume procurement drive scale.

SYNTHESIS

Beyond the big three

THE GULF: capital

Sovereign money, not models. Stargate UAE (5 GW; G42, MGX), Saudi HUMAIN, and MGX committed up to $500B to US Stargate. Abundant capital and compute, thin domestic talent and market.

THE GLOBAL SOUTH: the battleground

Where US and Chinese open models compete for the next billion users. Sovereign-language models (SEA-LION, Sahabat, TAIDE) run on rented compute, increasingly on Chinese open weights.

Three hubs build and govern AI. Everyone else deploys it, increasingly on Chinese open weights, which is itself a geopolitical outcome.

The master comparison

Scope	United States	Europe / EU	Asia (China-led)
Governance	Market-led, light federal + states	Comprehensive AI Act (softening)	State-driven (China); mixed across Asia
Frontier models	Dominant, closed	Thin (Mistral; DeepMind under Google)	Open-weight (China); KR/JP/IN building
Capital	~83% of global private AI $	Late-stage capital gap	State guidance funds; Gulf money
Compute	Power-constrained	One chokepoint (ASML)	Owns the hardware: TSMC, HBM, Foxconn
Deployment	Enterprise SaaS	Compliance-gated	Super-app, population scale
Clinical AI	Most approvals, underpaid	Most regulated	Most deployed (China); Korea exports
Founder-friendly	Capital + speed; talent cost up	Subsidy + certainty; heavy rules	State capital (China); open (India/SEA)

Takeaways and outlook

01 No region does all three well. The US builds the frontier, China deploys at scale, the EU regulates, and each is winning a different race.

02 The structural divergence is open vs closed. China's open-weight surge is reshaping who deploys AI across the Global South, not just at home.

03 Constraints differ by region: US power, EU capital + its own rules, China advanced chips, but China's domestic stack and energy surplus are closing the gap.

04 In medicine the same split holds: US approves most, EU regulates hardest, China deploys most. The deployed-AI bottleneck is institutional, not just technical.

References

Topic	Source
US AI Action Plan; preemption EO	White House (Jul 2025, Dec 2025)
EU AI Act + Digital Omnibus	White & Case; European Parliament (2025-26)
China GenAI + labeling rules	China-Briefing; Loeb (2023-25)
Chip controls; H20/H200; revenue-share	BIS; PBS; Tom's Hardware (2022-26)
China domestic chips; CloudMatrix	SemiAnalysis; Tom's Hardware (2025-26)
Adoption; ROI gap (95% pilots)	McKinsey; Eurostat; MIT/Fortune (2025)
Open-weight share; models; patents	MIT Tech Review; Stanford HAI (2025-26)
Investment; VC; valuations	Stanford AI Index; PitchBook (2025-26)
Talent: H-1B fee; China K-visa	Bulletin of Atomic Scientists; C&EN (2025)
Clinical AI: FDA / EU / NMPA	npj Digital Medicine; JMIR; Reed Smith (2025-26)

Full citations with URLs in the companion research.md.

Thank You

Curtis Chambers · Jinchi Wei

CPH 200C · Computational Precision Health

夢想